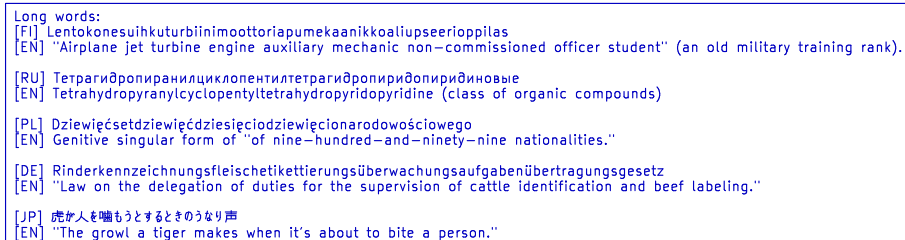


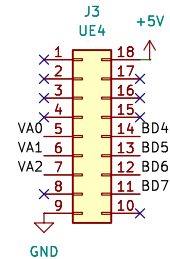
The diagram shows a 16-bit RAM module with three main components: J1 (RAM Decoder), U2 (74HCT138), and U4 (74HCT30). J1 is a 16-pin connector with pins 1-8 labeled VA10, VA11, VA12, BLK0/G2A, VA13/G1, and pins 9-16 labeled GND. U2 is a 16-pin 3-to-8 decoder with pins 1-3 labeled A0, A1, A2, pins 4-6 labeled E0, E1, E2, and pins 7-16 labeled Y0-Y7. U4 is an 8-input NAND gate with pins 1-8 labeled RAM0-RAM7 and pin 9 labeled RAM_CS. The circuit is powered by +5V and grounded to GND.

Spare Gates

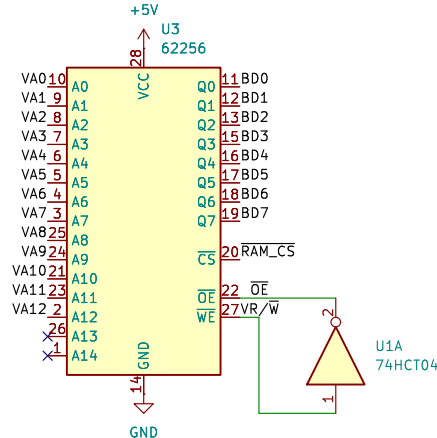


Pinout diagram for J2 connector:

Pin	Signal
1	VA6
2	VA5
3	VA4
4	VA3
5	BD0
6	BD1
7	BD2
8	BD3
9	GND
10	VR/W
11	BD3
12	BD2
13	BD1
14	BD0
15	VA9
16	VA8
17	VA7
18	+5V



The diagram shows two 74HCT00 inverters, U1G and U4B, connected to a common bus. Each inverter has its VCC pin connected to +5V and its GND pin connected to GND. The output of U1G (pin 14) is connected to the input of U4B (pin 7). The output of U4B (pin 14) is connected to the input of U1G (pin 7). The common bus is connected to +5V and GND. Decoupling capacitors C1, C2, C3, and C4 are connected between the bus and GND. C1 and C2 are 100nF capacitors, and C3 and C4 are 100nF capacitors.



Read: $\overline{CS}$ low & $VR/\overline{W}$ high $\rightarrow \overline{OE}$ low
Write: $\overline{CS}$ low & $VR/\overline{W}$ low $\rightarrow \overline{OE}$ high

U1	74HCT04 Hex Inverter
U2	74HCT138 3-to-8 Decoder
U4	74HCT30 8-Input NAND
C1 to C4	100nF Decoupling Capacitor
J1	2x8 Dupont Female-Female Cables to VIC
J2 & J3	Pair of 1x9 SII Pin Headers or 2x9 Wirewrap Sockets

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